Command Broker Framework

Cross-Sector Human Interposition Requirements

Applicable Across Critical Infrastructure and Democratic Governance Domains

Companion Framework to the DBA Sector Series and the Democratic Safeguards Project

Version 0.2 — DRAFT

May 2025 · Revised July 2026

Tim Roxey

Eclectic Technologies

For review by sector regulators, AI governance bodies, and critical infrastructure policy stakeholders

1. Purpose and Scope

This document establishes a cross-sector framework for the Command Broker function — the human interposition requirement applicable wherever a Generative AI (Gen AI) system's output can produce a consequential action in a critical process, physical or otherwise. It defines the broker concept, the conditions under which it applies, the architectural requirements that give it force, and the qualification criteria that determine who may fill the role.

The framework is organized at the cross-sector level because the broker requirement is not sector-specific. It derives from a sector-independent property of Gen AI systems: they are non-deterministic. The same prompt can produce different outputs across execution cycles. Formal verification methods that bound the behavior of deterministic systems do not apply. In any domain where a Gen AI output can initiate or materially influence a consequential action — whether that action is a change to a physical process parameter or the publication of content intended to influence political opinion — the same functional requirement applies: a qualified human must be interposed between the Gen AI output and the consequential action, and that interposition must be enforced by architecture, not by procedure or operator discipline alone.

Two applications of this requirement are currently being developed within the Eclectic Technologies body of work. The Command Broker governs Gen AI deployment in critical infrastructure industrial control systems, where the consequential action is a change to a physical process parameter. The Influencer Broker governs Gen AI deployment in the information environment, where the consequential action is the publication or amplification of content capable of influencing political or social outcomes at scale. Both are developed in this framework. Both are derived from the same source requirement: the non-determinism of Gen AI and the consequential nature of its outputs.

Sector-specific implementation details — consequence thresholds, band assignments, and qualification standards adapted to the sector's operational context — are set out in sector annexes to this framework. The annexes currently in development or anticipated are: DBA-W (water sector), DBA-GAS (natural gas transmission), DBA-OIL (petroleum sector), DBA-Nuclear (nuclear facilities), and the Democratic Safeguards project (information environment and democratic governance). The cross-sector framework establishes the architecture and qualification philosophy; the annexes apply them.

2. The Broker Concept — A General Principle

The broker is a human decision-maker who stands between an AI system’s recommendation and the consequential action it would initiate or influence. The broker’s function is not to relay the AI’s recommendation. It is to evaluate the recommendation against independent operational knowledge, current conditions, and professional judgment, and to authorize, modify, or reject the recommended action before it takes effect. The broker exercises genuine decision authority. The interposition is substantive, not ceremonial.

The broker concept is necessary because of a specific property of Gen AI that distinguishes it from the machine learning (ML) systems that preceded it in industrial and information environments. ML systems — anomaly detection, predictive maintenance, pattern recognition algorithms — are deterministic within bounded input ranges. Given bounded inputs, they produce bounded, testable outputs. Their behavior can be characterized through semi-formal methods. They can, under appropriate conditions and with appropriate oversight, operate autonomously.

Gen AI systems, including large language models and related architectures, are not deterministic in this sense. The same input can produce meaningfully different outputs across execution cycles. The statistical processes that generate Gen AI outputs do not lend themselves to formal verification by methods that bound ML behavior. This is not a temporary limitation of current architectures that better engineering will resolve. It is a structural property of systems that generate novel outputs from learned distributions. That property does not make Gen AI unsuitable for deployment in consequential domains — but it does make unmediated autonomous action by Gen AI systems unsuitable wherever the consequences of error are significant.

The broker requirement is the response to that structural property. It is derived from the author's analysis of ML and Gen AI risk in critical infrastructure industrial control systems, refined through the ISAGCA Supplier Ad-hoc Working Group, which he convened and moderated from its first working session in February 2024. ISA published a snapshot of that analysis (Version 20) in 2025; the work has continued independently since and appears in its current edition as "Generative AI Risk in Industrial Control Systems" (Version 29, Eclectic Technologies, 2026). That work establishes a three-tier criticality framework applicable across all critical infrastructure sectors. This framework extends the broker concept beyond the industrial control system domain to encompass the information environment, where Gen AI's non-determinism poses a structurally equivalent risk.

Mechanism and accountability. The broker requirement has two parts that must not be conflated. The first is the enforcement mechanism — the means by which a consequential action is held until the interposition is satisfied. Where a consequence reduces to a checkable physical or operational constraint, that mechanism is implementation-neutral: an interlock, a hard limit, or a verified deterministic component may perform the enforcement, and does so more reliably than operator discipline. The second is the accountable authority — the party that holds the duty of care, answers to a reviewer, a regulator, or a board of inquiry for the decision, and whose judgment the audit record exists to make defensible. Accountability is not a property that added constraints confer; it requires an entity that can bear an obligation and be answerable for an outcome. A non-deterministic AI system cannot be that entity, and neither can a deterministic mechanism.

It follows that the question of whether a sufficiently constrained AI system could satisfy the broker criteria is answered by this distinction rather than by a prohibition. An AI system, or an automated component, may implement the enforcement mechanism where the consequence is checkable; it may not be the broker, because the broker is the accountable authority, and that authority remains with a named human decision-maker or governance body. Where a consequence cannot be reduced to a deterministic check — an integrity failure, an influence effect, or an accurate recommendation framed to induce an unsafe decision — accountability is non-delegable, and the interposition must be discharged by a human broker whose reasoning is preserved in the decision record. This principle applies equally to the Command Broker and the Influencer Broker defined below.

3. The Command Broker

3.1 Definition

The Command Broker is the human operator who decides — who brokers — a course of physical action based on process observables and advisory output from a component within scope of the Bright Line. The Command Broker function exists to ensure that no recommendation from an in-scope component — a generative or otherwise non-deterministic component, as defined by the scope carve-out in FD-LD §3 and FD-BL §4.4 — changes a physical process parameter, where that change is an above-line action under FD-BL §4.2, without a human decision interposed between the advisory output and the physical action. Deterministic machine learning within a stated input bound is carved out of this obligation (FD-LD §3); it is not carved out because it is machine learning but because its behaviour is bounded by proof.

The Command Broker is not a procedural role that can be satisfied by policy language, training certification, or workflow documentation alone. The Command Broker architecture requires that the control system enforce the broker function mechanically: the system must be incapable of executing an above-line physical process command originating from a component within scope of FD-BL §4.4 without a positive human confirmation step that is local, auditable, and not itself executable remotely or by another software component.

3.2 The Bright Line

The Bright Line is defined in FD-BL — Foundational Definition — The Bright Line, which governs. It is a threshold on actions, not a boundary between architectural layers: an action is above the line when its incorrect performance could produce a consequence at or above the facility's severity threshold (FD-BL §4.1–§4.2). An above-line action may not execute without passing through a human Command Broker, and that interposition is enforced by architecture, not by procedure. The advisory-to-command seam is a useful place to build that enforcement; it is not the definition of the line, and a single gate placed there on the assumption that it coincides with the line will leave above-line actions ungated on the advisory side and gate below-line actions needlessly on the command side (FD-BL §4.2). A vendor's characterization of embedded generative AI as “autonomous process optimization” or “direct control integration” is a product description, not a compliance determination; placement follows what the component's actions can cause. The Design Basis requirement governs; the vendor characterization does not.

The Bright Line is based on consequence analysis, not on system sophistication. The relevant question is not how capable the Gen AI system is, but what happens if it produces an erroneous output that is acted upon. A simple AI model that recommends a chemical dosing adjustment at a water treatment plant is above the Bright Line because its erroneous output could lead to a public health consequence. A sophisticated AI system performing energy optimization in a data center may be below the Bright Line if the consequence of its erroneous output does not reach the defined consequence threshold. Sophistication does not govern. Consequence governs.

§3.2a Placement is determined action by action. The unit of placement is the individual action, not the system or the application. A single generative component deployed once in one facility will typically produce actions on both sides of the line — e.g. drafting a maintenance work order (below), recommending a setpoint change during operation (placed on what an incorrect change could cause). The required architecture is therefore a consequence classifier that routes action by action, not a single gate at a seam; the classifier's routing decisions and their basis are part of the evidentiary record.

Placement is distinct from criticality-band assignment (§3.3). Band assignment is a property of the application, derived from its consequence-severity profile, and sets the rigour of assessment, qualification, and audit. Placement is a property of each action and sets whether that action may proceed without human authorization. A Highest-band application contains below-line actions; a Lowest-band application may contain above-line ones. Neither determines the other.

3.3 Criticality Bands

Applications are assigned to one of three criticality bands — Highest, Mid, Lowest — by the severity of the worst credible consequence of a cognitive error, measured across the six consequence categories and aggregated by taking the maximum (see the Consequence-Severity Tier Model, which is the normative basis for this assignment, and §5.3 below). The band for any specific application is fixed by consequence analysis in the applicable sector annex. The bands are:

Highest Criticality. The most rigorous assessment, qualification, and audit regime applies. Whether any specific action within such an application may proceed autonomously is decided by its placement under FD-BL §4.2 (above-line actions require a Command Broker; below-line actions do not), not by whether the producing component is ML or Gen AI. Determinism decides only whether a component is in scope of Bright Line analysis (FD-LD §3 / FD-BL §4.4).

Mid-Level Criticality. Gen AI informs; human decides. Gen AI may present recommended actions; the operator decides which action to execute. A human remains in the command loop. ML may operate autonomously.

Lowest Criticality. Gen AI may operate as a virtual assistant, taking limited autonomous actions where consequences do not affect critical functions or safety. ML operates with full autonomous behavior.

Naming crosswalk (legacy reading aid). The band names used here are canonical. Earlier documents in the corpus, and the ISA Parent Framework, numbered the same bands Tier 1 / 2 / 3; that numbering is retained below only to read older text. The parenthetical type-words that once accompanied the numbers — safety-critical, quality-of-service, administrative — are not band definitions; they name error classes (assessed at any band) and are defined as such in the Tiered Assessment Framework. New work uses the band names.

Criticality band (canonical)	Legacy number	Severity — worst credible consequence, any category
Highest	Tier 1	Design-Basis level: population- or mission-scale, loss of a critical function, or life-safety
Mid	Tier 2	Significant but bounded and recoverable, below the Design-Basis threshold
Lowest	Tier 3	No reach to a critical function, life-safety, or critical service in any category

3.4 Command Broker Qualification

The qualification criteria for a Command Broker must be derived from the operational context of the sector and facility in which the broker function is performed. The cross-sector framework establishes the qualification methodology; sector annexes define the specific criteria. The methodology requires that the following dimensions be addressed in each annex:

- Technical competency. The Command Broker must possess sufficient technical knowledge of the physical process to evaluate an AI recommendation independently — not to replicate the AI’s analysis, but to recognize when the recommendation is inconsistent with current conditions, operating constraints, or applicable limits. A Command Broker who accepts AI recommendations without independent evaluation is not performing the broker function.
- Process authority. The Command Broker must hold formal authority to authorize or reject changes to the relevant process parameters. The broker role cannot be assigned to an observer, an advisor, or a position that requires escalation before acting. The broker decides.

- Situational awareness. The Command Broker must have real-time access to the process observables needed to evaluate AI recommendations in context. A broker who can see only the AI's recommendation, without the underlying process data the recommendation is based on, cannot perform an independent evaluation.
- Adversarial stress competency. The Command Broker must be qualified to perform the broker function under degraded conditions — loss of normal communications, partial loss of sensor feeds, elevated operational tempo, or conditions arising from a military action or cyberattack scenario. Broker qualification programs must include training and evaluation under degraded conditions, not only under normal operating assumptions. This dimension is addressed in detail in Section 3.6.

3.5 Audit and Verification

Architectural enforcement of the broker function is necessary but not sufficient. The Command Broker must demonstrably be in the loop, not nominally so. Audit requirements must address three questions:

- Is the broker confirmation step being performed before physical commands execute, or is it being bypassed through workarounds, pre-authorization, or system configuration that effectively bypasses the enforcement mechanism?
- Is the broker's confirmation substantive — reflecting an independent evaluation of the AI recommendation — or is it a pro forma acknowledgment? Audit records must capture sufficient information to distinguish between genuine evaluation and rubber-stamping.
- Is the broker qualification maintained current? An individual qualified at the time of system deployment who has not maintained technical competency as the process or the AI system has changed is not a qualified broker. Qualification must be maintained, not merely achieved.

3.6 Command Broker Function Under Stress Conditions

The Design Basis Accident — Military Action (DBA-MA) framework and the DBA sector series establish that the Design Basis for critical infrastructure must account for conditions more severe than normal operation, including deliberate adversarial action. The Command Broker qualification framework must reflect this. A broker qualified only for normal operating conditions is under-qualified for the environments in which the broker's function is most consequential.

Stress conditions relevant to Command Broker qualification include: loss of external communications requiring the broker to act on local information alone; partial degradation of sensor feeds requiring the broker to identify and compensate for missing or unreliable data; elevated operational tempo in which multiple recommendations require evaluation in rapid sequence; and conditions in which the AI system itself may be operating from compromised or manipulated inputs — the cyberattack scenario in which the AI's recommendation is itself adversarially influenced.

The qualification program for Command Brokers at the Highest-band facilities must include scenario-based evaluation for each of these stress categories. Sector annexes specify the scenario content appropriate to their operational environments.

4. The Influencer Broker

4.1 Definition

The Influencer Broker is the human decision-maker who stands between a Gen AI system's content output and its publication or amplification to an audience that can be influenced at scale. The Influencer Broker's function is to evaluate AI-generated content for accuracy, contextual appropriateness, and potential to distort political or social outcomes, and to authorize, modify, or reject it before it reaches an audience.

The Influencer Broker applies in any deployment context where Gen AI is used to generate content — text, image, audio, or synthetic media — that is intended to inform, persuade, or mobilize an audience, and where that audience is sufficiently large or strategically positioned that the content's effects can be consequential at a societal level. The Influencer Broker concept was developed in the Democratic Safeguards project. This section establishes its relationship to the Command Broker and its common derivation from the broker principle.

4.2 The Common Derivation

The Command Broker and the Influencer Broker are the same functional requirement applied to different consequence domains. The structural parallel is precise.

In the Command Broker context, a Gen AI system produces a recommendation; if acted upon without review, the recommendation could change a physical process parameter, potentially causing physical harm. The broker interposes between the recommendation and the physical action.

In the Influencer Broker context, a Gen AI system produces content; if published without review, that content would enter the information environment, potentially distorting political or social outcomes. The broker interposes between the content and its publication.

The parallel holds at the level of mechanism, not merely at the level of analogy. In both cases, the Gen AI system is non-deterministic and cannot be formally verified; the output is consequential; the consequence cannot be recalled once the action is taken; and the only reliable protection against erroneous or manipulated output producing a consequential result is a qualified human evaluation before the action is completed. The Bright Line concept applies equally — the broker enforces the boundary between the AI advisory layer and the consequential action, whether physical or informational.

4.3 On the Question of Politicization

The inclusion of the Influencer Broker in this framework alongside the Command Broker will, for some readers, raise the question of whether a document that addresses both industrial control system requirements and democratic governance requirements has become a political document. That concern is understandable, and it is worth addressing directly.

This framework is not a political document. The inclusion of the Influencer Broker is not a statement about any political party, candidate, government, or ideology. It is a statement about a tool's properties.

The tool at issue is Gen AI — specifically, large language models and related architectures that generate text, images, and synthetic media from learned statistical distributions. That tool is being deployed across industrial control systems and information environments simultaneously. The same model family that a pipeline operator might use to recommend valve positions is being used by media organizations, political campaigns, and information operations to generate content at scale. The tool is the same. Its non-determinism is the same. Its incapacity for formal verification is the same. The functional requirement that a qualified human be interposed between its output and a consequential action is the same.

A framework that addressed the broker requirement in industrial control systems but not in the information environment would be analytically incomplete, not because the information environment is politically important, but because the same tool is operating in both environments and presenting the same structural risk in both. Addressing both in a unified framework reflects analytical rigor, not political alignment.

Sector regulators reading this framework in the context of critical infrastructure compliance are not being asked to adopt, endorse, or engage with any position on democratic governance. The Command Broker requirements applicable to their sectors are fully self-contained in this document and in the applicable sector annexes. The Influencer Broker material is provided for analytical completeness and because the qualification methodology — independent evaluation of AI output before consequential action, enforced by architecture rather than procedure — is shared. Regulators who wish to engage only with the Command Broker material may do so without loss of technical content.

5. Cross-Sector Architecture and Annex Structure

5.1 Relationship to the Sector Design Basis Series

Each sector in the DBA series establishes a Design Basis for the facilities within its scope. The Design Basis defines the envelope of credible events — including military action under the DBA-MA methodology — and the required protective architecture. Where that architecture includes ML or Gen AI components, the Command Broker requirement applies.

The sector Design Basis documents carry the Command Broker as an architectural requirement: the Bright Line must exist; Gen AI may not operate autonomously above it; a qualified human Command Broker must be in the control loop for every change to a physical process parameter above the Bright Line. The Design Basis does not include implementation details. Implementation detail — sector-specific consequence thresholds, band assignments, qualification standards, integration-pattern guidance, and audit criteria — is carried in the sector-specific implementation annexes to this framework.

5.2 Current and Anticipated Annexes

The following sector annexes are currently in development or anticipated:

- DBA-W Annex — Water Sector. Consequence thresholds: population-scale public health impact, loss of wastewater containment, loss of service to critical dependent facilities. Key operational context: SCADA-integrated AI, legacy system constraints, Oldsmar 2021 reference event. Qualification standard: water treatment and distribution system operations competency.
- DBA-GAS Annex — Natural Gas Transmission. Consequence thresholds: uncontrolled release events, electricity-gas interdependency cascade, cross-border supply disruption. Key operational context: compressor station automation, pipeline hydraulics, linepack management. Qualification standard: gas transmission operations and pipeline control competency.
- DBA-OIL Annex — Petroleum Sector. Consequence thresholds: process excursion producing fire, explosion, or hazardous material release; refinery multi-product supply disruption. Key operational context: refinery distributed control systems, process unit interdependencies. Qualification standard: refinery process control and safety system competency.
- DBA-Nuclear Annex — Nuclear Facilities. Consequence thresholds: reactor protection system challenge, spent fuel thermal progression, and radiological release. Key operational context: defense-in-depth architecture, safety system independence requirements, DBA-MA military action stress conditions. Qualification standard: licensed reactor operator competency with AI-system evaluation training. Note: this annex does not yet exist as a drafted document; it is anticipated as the nuclear sector AI governance work matures.
- Democratic Safeguards Annex — Information Environment. Consequence thresholds: content capable of distorting political outcomes at scale, synthetic media designed to undermine electoral processes, and AI-generated information operations targeting

democratic institutions. Qualification standard: editorial judgment, source verification, and AI content evaluation competency. Developed under the Democratic Safeguards project.

5.3 Consequence-Derived Band Assignment

Each sector annex must assign its applications to the criticality bands through a consequence analysis specific to the sector's operational context. The cross-sector framework establishes the methodology; the annexes provide the analysis. Band assignment is not transferable across sectors without independent analysis. An application that falls below the Bright Line in one sector may fall above it in another, depending on the consequence environment.

The consequence analysis scores the application in each of the six consequence categories it can reach — public health & safety, physical, environmental, economic & service-reliability, national security, and democratic & societal — nil in any it cannot. Within each category, severity is set by three modifiers: scale (facility, regional, national, societal), reversibility (recoverable within a defined timeframe vs. not), and conditions (normal, degraded, adversarial stress — the DBA-MA envelope). The application's band is the maximum severity reached in any single category; the category that sets it — the pinning category — is recorded, because it also identifies the governing regulatory regime. This is the aggregation rule defined in the Consequence-Severity Tier Model; the annexes supply the per-category thresholds.

6. Foundational References

Roxey, T.E. "Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers." Version 29. Eclectic Technologies, 2026. [Primary source for the three-tier criticality framework and the Bright Line concept. This edition supersedes the prior ISAGCA-attributed "A Study of AI Risks in Critical Infrastructure — Vendor Agnostic," Version 28.]

Roxey, T. DBA-W-Sector_Framework_v1_0. Eclectic Technologies. [Water sector Design Basis establishing the Command Broker as a sector Design Basis requirement and developing the Section 5.5 command-broker architecture.]

Roxey, T. TAF-W_AI_QA_Water_Sector_Guide_v1_2. Eclectic Technologies. [Water sector implementation guide for Command Broker architecture and deployment.]

Roxey, T. CB-CT Commissioning and Joint Training (CB-CT_Commissioning_and_Joint_Training v0.2). Eclectic Technologies, 2026. [Companion module sequencing the vendor–AOO handoff for commissioning and joint training of the Command Broker.]

Democratic Safeguards Project. [In development. Source for the Influencer Broker concept and the democratic governance application of the broker principle.]

Revision History

Version	Date	Description
0.1	May 2025	Initial draft. Establishes cross-sector broker concept, Command Broker and Influencer Broker definitions, three-tier criticality framework, qualification methodology, and annex structure. Addresses the relationship between the two broker types and the politicization question.
0.1 (housekeeping)	June 2026	Housekeeping correction. “Brightline” (one word) corrected to “Bright Line” (two words, title case) throughout all body text, section headings, and references, consistent with series-wide convention established in DBA-EN-SF §4.3. No substantive content changes.
0.2	Jul 2026	Reconciled to the Consequence-Severity Tier Model. §3.3 recast as criticality bands (Highest/Mid/Lowest) on the six-category take-the-max basis; naming crosswalk demoted to a legacy reading aid and the type-words (safety-critical/quality-of-service/administrative) withdrawn as band definitions and returned to the error classes; §5.3 promoted to the six-category consequence basis with the pinning-category rule; band vocabulary applied to the framework's own normative text. Descriptions of the fixed source study (V29) left verbatim.
0.2 (cite refresh)	Jul 2026	Citations refreshed to current corpus versions: CB-Framework v0.2, Command Broker Implementation Guide V0.2, CB-CT v0.2, DBA-VC v0.8. Mechanical version-reference pass; no substantive content change. (Held pending decision: the NERC/BES-Variant TAF reference and the ISA Parent Framework V1.2 label.)